

# **Social Media Text as Data**

## **Exploring the Potential and Limitations of Automated Content Analysis Methods**

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## Section 1

### **Part I: Why Social Media Text Matters**

# The Data Revolution in Marketing Research

## Traditional data sources relied on:

- Surveys and focus groups (slow, expensive, self-reported)
- Sales and transaction records (behavioral, but limited)
- Panel data (structured, but retrospective)

## Social media changes everything:

- **Scale:** Billions of posts daily
- **Speed:** Real-time signal
- **Naturalistic:** Unsolicited opinions
- **Richness:** Emotional, contextual
- **Accessible:** APIs, scrapers
- **Longitudinal:** Historical archives

*Source: Miller (2015), Ch. 10 | Jimmy Zhenning Xu, Ph.D. | [github.com/utjimmyx](https://github.com/utjimmyx)*

# Platforms and Their Data Ecosystems

| Platform  | Primary Format   | API Access          | Best For           |
|-----------|------------------|---------------------|--------------------|
| Twitter/X | Short text       | Restricted (v2)     | Public opinion     |
| Reddit    | Long-form        | Free (pushshift)    | Niche communities  |
| Bluesky   | Short text       | Open (AT Protocol)  | Emerging discourse |
| YouTube   | Comments         | YouTube Data API v3 | Product feedback   |
| LinkedIn  | Professional     | Limited             | B2B signals        |
| TikTok    | Video + captions | Restricted          | Gen Z trends       |

*Each platform requires platform-specific scraping strategy and ethical considerations.*

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# What Can We Learn from Social Media Text?

## Key constructs researchers extract:

- 1 **Sentiment** — positive, negative, neutral valence
- 2 **Topics** — what issues are being discussed?
- 3 **Emotions** — anger, joy, fear, anticipation (NRC)
- 4 **Brand associations** — co-occurring terms and entities
- 5 **Behavioral signals** — purchase intent, complaints, advocacy
- 6 **Network structure** — who talks to whom, influence flow

## Key Question

Can machines reliably measure these constructs at scale — and when do they fail?



## Section 2

# **Part II: Core Methods of Automated Text Analysis**



# The Automated Content Analysis Toolkit

*# Conceptual overview - not executable*

```
methods <- list(  
  lexicon_based = c("AFINN", "Bing", "NRC", "VADER"),  
  ml_supervised = c("Naive Bayes", "SVM", "Random Forest"),  
  deep_learning = c("BERT", "RoBERTa", "LLaMA"),  
  topic_models  = c("LDA", "STM", "BERTopic"),  
  classical_nlp  = c("TF-IDF", "n-grams", "Named Entity Recognition")  
)
```

| Method Family       | Examples                        |
|---------------------|---------------------------------|
| Lexicon-based       | AFINN, Bing, NRC, VADER         |
| ML (Supervised)     | Naive Bayes, SVM, Random Forest |
| Deep Learning / LLM | BERT, RoBERTa, GPT-4            |
| Topic Models        | LDA, STM, BERTopic              |
| Classical NLP       | TF-IDF, n-grams, NER            |

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# Lexicon-Based Sentiment Analysis in R

**tidytext** pipeline with multiple lexicons:

```
library(tidytext)
library(dplyr)

# AFINN: integer scores -5 to +5
afinn_scores <- tokens_df |>
  inner_join(get_sentiments("afinn"), by = "word") |>
  group_by(post_id) |>
  summarise(sentiment_score = sum(value))

# Bing: binary positive/negative
bing_counts <- tokens_df |>
  inner_join(get_sentiments("bing"), by = "word") |>
  count(post_id, sentiment) |>
  pivot_wider(names_from = sentiment,
              values_from = n, values_fill = 0) |>
  mutate(net = positive - negative)
```

# Topic Modeling: Latent Dirichlet Allocation (LDA)

**Core intuition:** Documents are mixtures of topics; topics are distributions over words.

```
library(topicmodels)
library(tidytext)

# Create document-term matrix
dtm <- tokens_df |>
  count(post_id, word) |>
  cast_dtm(post_id, word, n)

# Fit LDA with k topics
lda_model <- LDA(dtm, k = 5,
                 control = list(seed = 580))

# Extract top terms per topic
lda_model |>
  tidy(matrix = "beta") |>
```

# TF-IDF: Identifying Distinctive Vocabulary

**Term Frequency–Inverse Document Frequency** identifies words that are important to a document *relative to a corpus*:

$$\text{TF-IDF}(t, d) = \underbrace{\frac{f_{t,d}}{\sum_k f_{k,d}}}_{\text{TF}} \times \underbrace{\log \frac{N}{|\{d : t \in d\}|}}_{\text{IDF}}$$

```
brand_tfidf <- tokens_df |>
  count(word) |>
  bind_tf_idf(word, brand, n) |>
  group_by(brand) |>
  slice_max(tf_idf, n = 10) |>
  ggplot(aes(tf_idf,
             reorder_within(word, tf_idf, brand))) +
  geom_col(fill = "#003D79") +
  facet_wrap(~brand, scales = "free")
```



## Section 3

# **Part III: Potential — What Automated Methods Do Well**

# Scale and Speed: The Core Advantage

## By the Numbers

A trained human coder can analyze ~**200–400** social posts per hour.  
An R pipeline can analyze **200,000+** posts per minute.

## Real-world applications where scale matters:

- Brand tracking across millions of mentions
- Crisis detection (sudden sentiment shift)
- Competitive intelligence monitoring
- Policy event impact analysis (e.g., FDA announcement → Reddit posts)
- Election forecasting via search & social volume

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**Automated methods offer key scientific advantages:**

## Human Coding

- Inter-rater disagreement
- Coder fatigue effects
- Hard to fully replicate
- Training drift over time

## Automated Methods

- 100% consistent rules
- Codebook = the code itself
- Fully reproducible
- Versioned, auditable

## Caveat

Reproducibility  $\neq$  validity. A machine can be consistently *wrong*.

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# Longitudinal Tracking and Event Studies

## Combining sentiment with event timelines:

```
library(ggplot2)
library(lubridate)

# Plot daily sentiment around product launch
sentiment_daily |>
  ggplot(aes(x = date, y = mean_sentiment)) +
  geom_line(color = "#003D79", linewidth = 1) +
  geom_ribbon(aes(ymin = ci_low, ymax = ci_high),
            alpha = 0.2, fill = "#003D79") +
  geom_vline(xintercept = as.Date("2024-09-15"),
            linetype = "dashed",
            color = "#DC5032") +
  annotate("text", x = as.Date("2024-09-15"),
            y = max(sentiment_daily$mean_sentiment),
            label = "Product Launch",
            hjust = -0.1, color = "#DC5032") +
```



## Section 4

# **Part IV: Limitations — Where Automated Methods Struggle**

# The Validity Problem

Are we measuring what we think we're measuring?

*# Consider these posts - all about the same brand:*

```
posts <- c(
  "This phone is sick!",           # Slang = positive
  "Not bad for the price",         # Negation + sentiment
  "I can't believe how good this is", # Double negation
  "Sure, if you like paying $1000 for a brick", # Sarcasm
  "My grandmother loves it "      # Emoji irony
)
# AFINN/Bing will misclassify most of these
```

## Common Failure Modes

- Slang & colloquialisms
- Sarcasm & irony
- Negation handling
- Domain-specific language

## Emoji & Multimodal

- Emoji sentiment reversal
- Memes with no text
- Code-switching languages
- Image-based humor

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## Lexicon Limitations: Context Blindness

**AFINN assigns fixed scores regardless of context:**

| Phrase                | AFINN Score | Actual Sentiment |
|-----------------------|-------------|------------------|
| "This diet is sick"   | -3          | Positive (slang) |
| "Not bad"             | -3          | Mildly positive  |
| "Killing it in sales" | -3          | Very positive    |
| "Surprisingly good"   | +3          | Positive         |
| "Could be worse"      | -2          | Mildly positive  |

### Key Limitation

Bag-of-words approaches discard word order, context, and pragmatics. Each word is treated as independent — a flawed assumption for natural language.

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# Sampling and Representation Bias

## Who is actually posting on social media?

- **Platform demographics** differ drastically from the general population
- **Vocal minority effect:** Most users lurk; a few post frequently
- **Bot contamination:** Estimated 5–15% of Twitter/X accounts are automated
- **Survivorship bias:** Deleted and private posts are invisible
- **Amplification algorithms:** What gets scraped  $\neq$  what was written

## Methodological Implication

Social media samples are **not random samples** of consumer opinion. Findings may not generalize to the broader market.  
(Jungherr et al., 2017; Hargittai, 2020)

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# The API Access Problem

Researcher access to social media data has eroded significantly:

- **Twitter/X (2023):** Free Academic API discontinued; basic tier: 500K tweets/month
- **Reddit (2023):** Third-party API costs up 10,000%; killed pushshift
- **TikTok:** No public research API; scrapers frequently broken
- **Meta:** CrowdTangle shut down (August 2024)

## Alternative Platforms

- **Bluesky:** Open AT Protocol (atrrr pkg)
- **Mastodon:** Open API
- **YouTube:** Still accessible via tuber
- **Reddit:** Limited free tier remains

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## Four key ethical dimensions (franzke et al., 2020):

### 1 Consent & Expectation of Privacy

Public  $\neq$  consented to research use. Context matters (Twitter vs. support groups).

### 2 Potential for Harm

Exposing vulnerable users, doxxing through data linkage, re-identification.

### 3 Intellectual Property

Platform ToS, copyright of user-generated content.

### 4 Algorithmic Bias

NLP models trained on biased corpora reproduce and amplify those biases.

*IRB review is required even for "public" social media data in most institutions.*  
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## Section 5

# **Part V: Best Practices and Validation**

# The Gold Standard: Human + Machine

## Recommended hybrid validation workflow:

```
# Step 1: Draw a stratified random sample
set.seed(580)
validation_sample <- posts_df |>
  slice_sample(n = 200, weight_by = abs(sentiment_score))

# Step 2: Export for human coding
write_csv(validation_sample, "for_human_coding.csv")

# Step 3: Compare machine vs. human labels
validation_results <- validation_sample |>
  left_join(human_codes, by = "post_id") |>
  mutate(agreement = machine_label == human_label)

# Step 4: Report agreement statistics
caret::confusionMatrix(
  factor(validation_results$machine_label)
```

# Choosing the Right Method for Your Research Question

| If you need...           | Use...               | Avoid...           |
|--------------------------|----------------------|--------------------|
| Quick sentiment at scale | AFINN / Bing         | LLM APIs (cost)    |
| Nuanced emotion          | NRC / VADER          | Simple polarity    |
| Topic discovery          | LDA / BERTopic       | Keyword lists      |
| Domain-specific terms    | Train custom lexicon | Off-shelf lexicons |
| High accuracy on small N | Fine-tuned BERT      | Bag-of-words       |
| Exploratory/unknown text | TF-IDF + topic model | Supervised ML      |

## Practical Rule

Start simple. Validate against human judgment. Escalate complexity only when simpler methods fail on your domain.

## Minimum reporting requirements (following Grimmer & Stewart, 2013):

- **Data provenance:** Platform, date range, API version, filters applied
- **Preprocessing decisions:** Stopwords, stemming/lemmatization, minimum frequency
- **Method justification:** Why this lexicon/model for this domain?
- **Validation evidence:** Agreement statistics (Cohen's  $\kappa$ , F1, accuracy) on held-out human-coded sample
- **Limitations disclosed:** Platform representativeness, known failure modes
- **Code availability:** Reproducible pipeline (GitHub preferred)

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## Section 6

# **Part VI: Emerging Frontiers**

# Large Language Models as Content Analysts

LLMs (GPT-4, Claude, LLaMA) offer new capabilities:

*# Zero-shot classification via Anthropic API*

`library(httr2)`

```
classify_sentiment <- function(text) {  
  req <- request("https://api.anthropic.com/v1/messages") |>  
    req_headers(  
      "x-api-key" = Sys.getenv("ANTHROPIC_API_KEY"),  
      "anthropic-version" = "2023-06-01"  
    ) |>  
    req_body_json(list(  
      model = "claude-sonnet-4-6",  
      max_tokens = 10,  
      messages = list(list(  
        role = "user",  
        content = paste0("Classify sentiment as positive,",  
                          " negative, or neutral. Reply with one word only.\n")
```



# LLMs: Promise and Peril for Social Media Research

## Advantages

- Handle sarcasm, irony, context
- Multilingual out-of-box
- Zero-shot for new tasks
- Nuanced emotion detection
- Structured output extraction

## Limitations

- Cost at scale (API pricing)
- Stochastic — not reproducible
- Model version drift
- Hallucination risk
- Black-box interpretability
- Data privacy concerns

## Research Frontier

Can we use LLMs to *validate* traditional methods rather than replace them? (Pangakis et al., 2023)

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# BERTopic and Neural Topic Models

## Moving beyond LDA: embedding-based topic discovery

Key advantages of neural topic models:

- 1 **Semantic coherence:** Topics based on meaning, not co-occurrence
- 2 **Dynamic topics:** Track how topics evolve over time
- 3 **Outlier handling:** Documents don't need to fit a topic
- 4 **No hyperparameter search:** HDBSCAN finds  $k$  automatically

```
# BERTopic via reticulate in R  
library(reticulate)  
bt <- import("bertopic")  
model <- bt$BERTopic(language = "english",  
                      calculate_probabilities = TRUE)  
topics <- model$fit_transform(docs)  
model$get_topic_info()
```



## Section 7

# **Part VII: Summary and Discussion**

# Key Takeaways

## What we covered today:

- 1 Social media text is a rich but **non-representative** data source
- 2 Lexicon-based methods are **fast and transparent**, but context-blind
- 3 Topic models reveal **latent structure** without supervision
- 4 LLMs offer **higher validity** but raise cost, reproducibility, and privacy concerns
- 5 **Validation against human judgment** is non-negotiable
- 6 **API access is fragile** — open platforms (Bluesky, YouTube) are your friends
- 7 **Ethics** must be built into the research design, not added at the end

## Xu's Rule

"Your AI is only as good as the data it is built on."

# Discussion Questions

## For reflection and class discussion:

- 1 A brand manager asks you to measure consumer sentiment using Instagram comments. What are the three biggest threats to validity in this study?
- 2 When would you prefer AFINN over a fine-tuned BERT model for a marketing research project? When would you switch?
- 3 A competitor appears to be flooding Reddit with positive posts about their product. How would automated methods detect — or miss — this?
- 4 How should researchers disclose the use of LLMs (e.g., GPT-4) as part of a content analysis pipeline in a peer-reviewed paper?

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# References

- Grimmer, J., & Stewart, B. M. (2013). Text as data: The promise and pitfalls of automatic content analysis methods for political texts. *Political Analysis*, 21(3), 267–297.
- Silge, J., & Robinson, D. (2017). *Text mining with R: A tidy approach*. O'Reilly Media. <https://www.tidytextmining.com>
- Pangakis, N., Wolken, S., & Fasching, N. (2023). Automated annotation with generative AI requires validation. *arXiv:2306.00176*.
- Jungherr, A., Rauchfleisch, A., & Posegga, O. (2017). Public opinion and online political communication. *Journal of Information Technology & Politics*, 14(1), 1–18.
- franzke, a. s., Bechmann, A., Zimmer, M., & Ess, C. (2020). *Internet research: Ethical guidelines 3.0*. AoIR. <https://aoir.org/reports/ethics3.pdf>
- Miller, T. W. (2015). *Marketing data science*. Pearson FT Press.

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